Super MuSR: Design, Stave Assembly, and Beamline Verification

Science and Technology Facilities Council

ISIS Neutron and Muon Source

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Introduction

Super MuSR is the cutting-edge upgrade of the current MuSR Instrument, increasing time resolution and the number of detector elements from 64 to 960 to yield a ~20x increase in data collection rates.

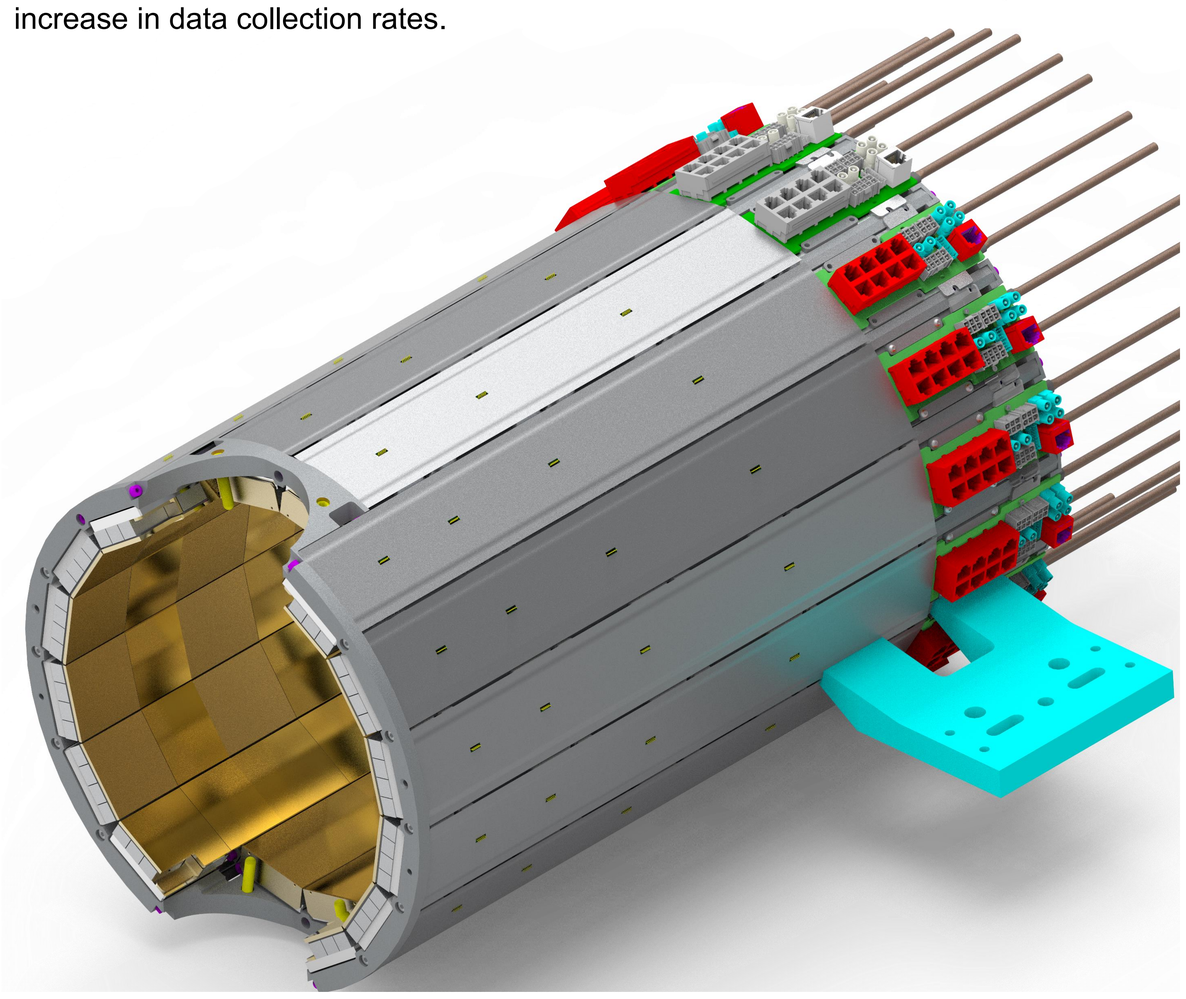


Figure 1: A Schematic of half of the Super MuSR Detector

Improved sample temperature and magnetic-field range will further extend capability for applications in superconductivity, complex/soft matter, quantum and battery technologies, and biological sciences [1].

Muon Delivery

The new beamline will deliver muons using a series of quadrupole triplets and dipole magnets to focus and direct polarised muon pulses. New spin rotators and a pulse slicer will be installed on the beamline [2].

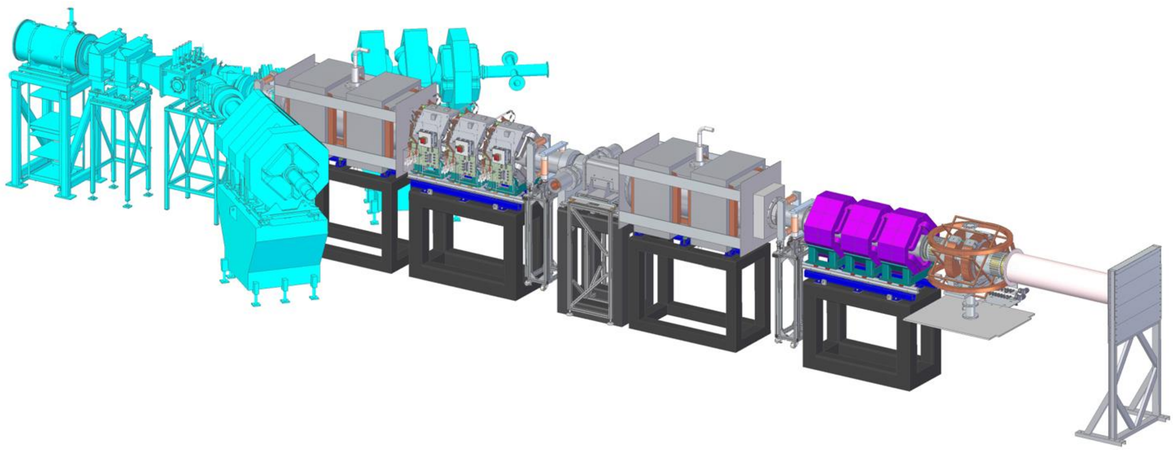


Figure 2: The Future Super MuSR beamline, in blue is the kicker that separates the three muon instruments, following are a set of quadrupole magnets, spin rotators and a pulse slicer before the muons arrive at the detector.

Detector Design

The detector uses compact, magnetically immune SiPMs (1.3 mm × 1.3 mm) and bespoke scintillating tiles to detect positrons and provide uniform, efficient light collection. The tiles, produced by RAL Technology, are made from scintillating plastic with wavelength-shifting fibre and clear epoxy resin. Tiles are then wrapped in ESR.

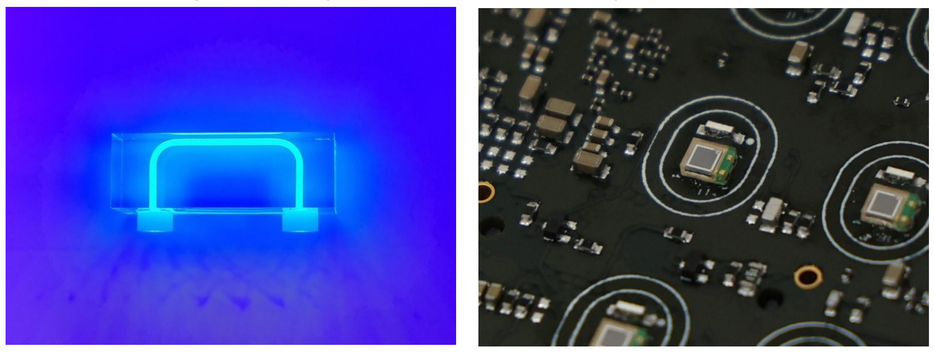
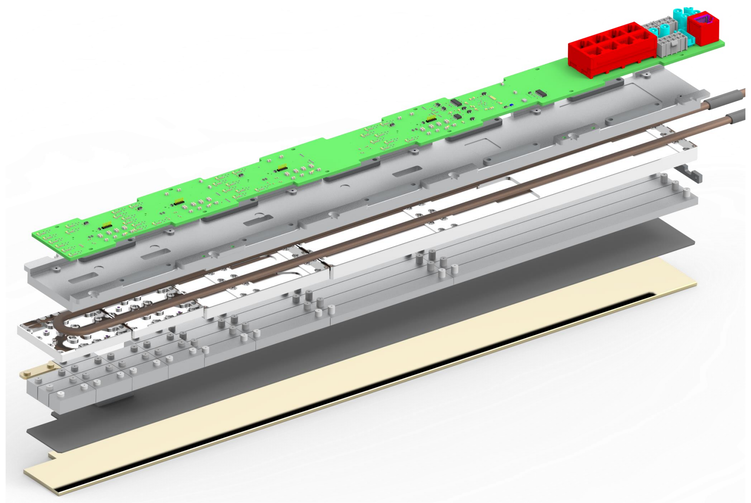


Figure 3: Unwrapped scintillating tile (left) and SiPMs (right). Optimisation of tile design included resin, temperature control, surface finishing, and fibre wrapping.

A single stave is constructed of 32 tiles arranged in groups of 8, tiles vary in length to conserve solid angle throughout all modules.



Modules are assembled by slotting the tiles into precision, tightly toleranced mechanical fixtures to ensure accurate alignment of the WLSF with the SiPM active area. The components are coupled using index-of-refraction matching optical media.

Figure 4: Exploded view of a single Super MuSR stave. Note: the two lower layers (not shown below) are a compression plate and a 3 mm brass degrader.

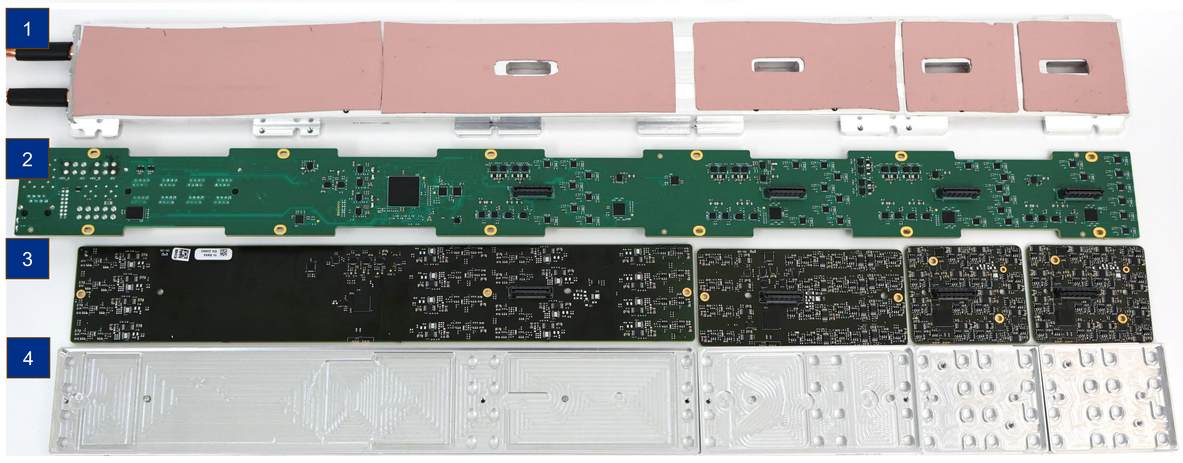


Figure 5: Mechanics and electronics required to assemble one stave. (1) Cooling plate. (2) Stave PCB. (3) Module PCBs of varying lengths. (4) Module Mechanics.

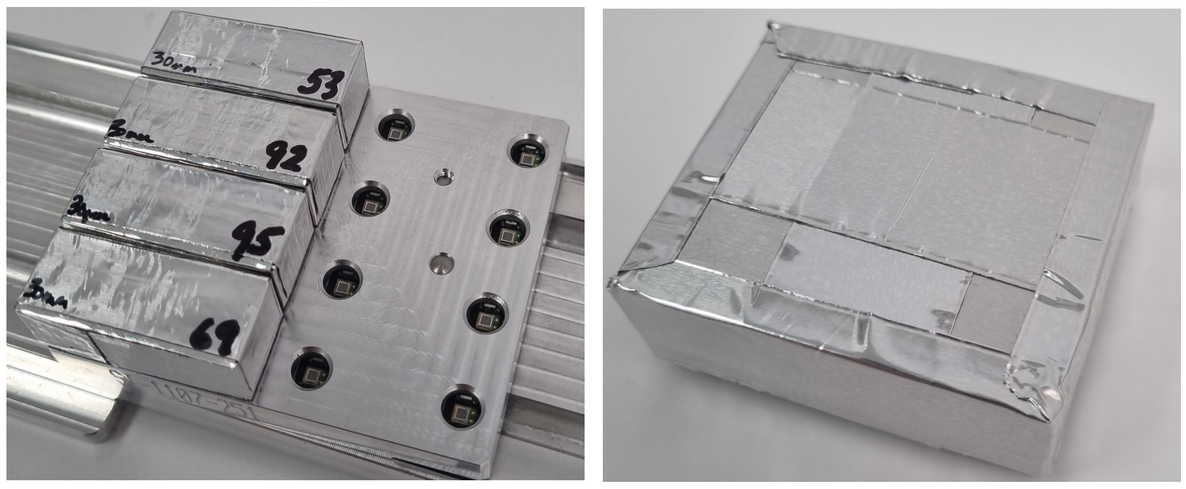


Figure 6: Example "Module A" assembly showing tiles slotted into mechanical fixtures (left). Tile studs are held flush to the SiPM with compression plates and aluminium tape, providing strong optical coupling and a light-tight seal (right).

Results

Muon pulses arrive from ISIS at 40 Hz and decay into positrons. The acquisition window is variable, but typically ~32 μs.

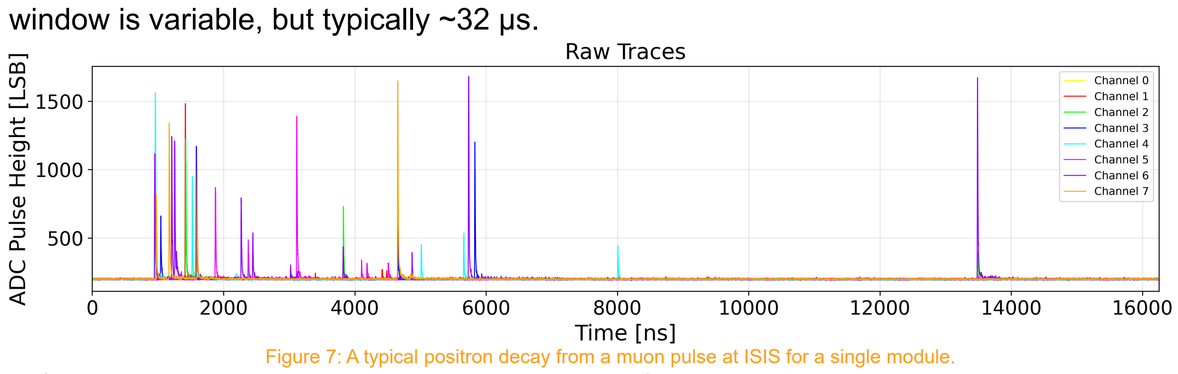


Figure 7: A typical positron decay from a muon pulse at ISIS for a single module.

A fixed-threshold discriminator is used to identify events. For each triggered event, the pulse height, timestamp, and channel/position are extracted. Events are accumulated into pulse height and time histograms.

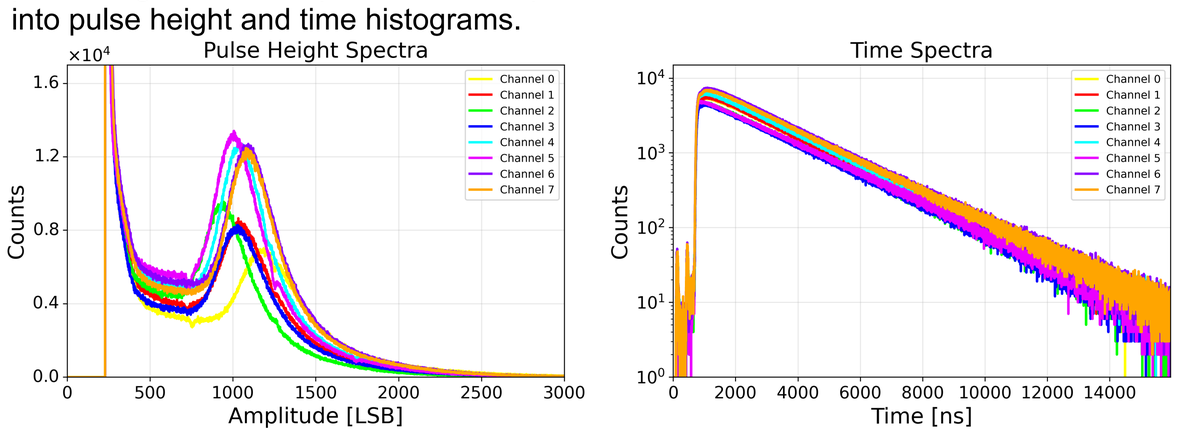


Figure 8: Example Pulse Height (left), and Time (right) Histograms.

Two production staves were assembled and tested on the current MuSR beamline. The assembly showcased excellent mechanical tolerances, including for the hand-made, labour-intensive scintillating tiles, with minimal breakage and variation in tile performance. During beamline measurements, the staves showed strong thermal stability. Detector light collection was also excellent: Ring 1 achieved a peak-to-valley ratio above 3, exceeding the Super MuSR performance requirements. Super MuSR will continue with full production and aims to have a complete detector test in September 2026.

References
[1] ISIS Neutron and Muon Source (2025) Super MuSR https://www.isis.stfc.ac.uk/Pages/Super-MuSR.aspx accessed: 07/02/2026
[2] P J Baker et al 2023 J. Phys.: Conf. Ser. 2462 012032